

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

Code of Conduct:

I D.Anand Paul(192425207) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

COMPARATIVE ANALYSIS OF NETWORK TOPOLOGIES

Introduction

Different network topologies have different characteristics in terms of cost, reliability, fault tolerance, complexity, scalability, performance, and maintenance.

The following comparative analysis evaluates Star, Bus, Mesh, and Hybrid topologies based on five important networking factors.

1. STAR TOPOLOGY VS BUS TOPOLOGY

1.1 Cost

Bus topology is generally less expensive than star topology because it traditionally uses a single backbone cable and does not require a central switch.

In a star topology, every device requires an individual connection to a central switch or hub. Therefore, more cabling and networking hardware are required.

Comparison

Factor	Star	Bus
Central device	Required	Not required
Cabling	More	Less
Initial cost	Higher	Lower
Hardware	Switch/hub required	Minimal

Therefore, bus topology has a cost advantage for very small traditional networks.

1.2 Reliability

Star topology is more reliable at the individual-node level.

If one computer or its connecting cable fails, only that computer is disconnected. The remaining devices can continue operating.

In a bus topology, failure of the main backbone can disrupt communication for all connected devices.

Therefore:

Star → Higher reliability

Bus → Lower reliability

1.3 Performance

Star topology generally provides better performance because modern Ethernet switches can provide dedicated links between devices.

In a traditional bus network, all devices share the same communication medium. As the number of devices and traffic increases, contention and collisions can increase.

Therefore, star topology is better suited for modern LAN environments.

1.4 Overall Comparison

Feature	Star	Bus
Cost	Higher	Lower
Reliability	High	Low
Performance	High	Lower
Troubleshooting	Easy	Difficult
Expansion	Easy	Difficult
Modern usage	Very common	Mostly obsolete

Conclusion

Although bus topology has a lower initial cost, star topology is generally preferred because it provides better performance, easier troubleshooting, and greater reliability.

2. MESH TOPOLOGY VS STAR TOPOLOGY – FAULT TOLERANCE

2.1 Mesh Topology

Mesh topology provides multiple communication paths between network devices.

In a full mesh, each node has a direct connection to every other node.

For example, if one link fails, traffic can potentially use another available link.

This provides excellent fault tolerance.

2.2 Star Topology

In star topology, all devices depend on a central switch or hub.

If one individual computer fails, other computers continue operating.

However, if the central switch fails, all devices connected through that switch may lose network connectivity.

Therefore, the central device represents a potential single point of failure.

2.3 Fault-Tolerance Comparison

Feature	Mesh	Star
Multiple paths	Yes	Usually no
Individual node failure	Usually tolerated	Tolerated
Central-device failure	No central dependency in full mesh	Can affect entire network
Fault tolerance	Very high	Moderate
Cost	Very high	Moderate
Complexity	High	Low

2.4 Example

Consider a network with four important devices.

In a mesh arrangement, multiple paths exist:

A ↔ B

A ↔ C

B ↔ D

C ↔ D

If one connection fails, another path may still be available.

In a basic star network, all devices depend on:

Device → Central Switch

If the central switch fails, communication between the connected devices stops.

2.5 Conclusion

Mesh topology provides greater fault tolerance than basic star topology because it provides multiple paths.

Therefore, mesh or partial-mesh designs are appropriate for critical environments such as data centers, telecommunications networks, and other high-availability systems.

3. BUS TOPOLOGY VS MESH TOPOLOGY – INSTALLATION COMPLEXITY

3.1 Bus Topology

Bus topology uses a shared backbone.

The traditional implementation requires:

- Backbone cable
- T-connectors
- Network connections
- Terminators

Because the devices share one main communication path, the physical structure is relatively simple.

3.2 Mesh Topology

Mesh topology is much more complex.

In a full mesh, every node must have a connection to every other node.

For n nodes, the number of links required in a full mesh is:

$$\text{Number of links} = n(n - 1) / 2$$

For example, if there are 5 nodes:

$$5(5 - 1) / 2 = 10 \text{ links}$$

For 10 nodes:

$$10(10 - 1) / 2 = 45 \text{ links}$$

This demonstrates how quickly complexity increases.

3.3 Comparison

Feature	Bus	Mesh
Installation	Simple	Complex
Number of links	Low	Very high

Cabling	Low	Very high
Configuration	Simple	Complex
Initial cost	Low	High
Scalability	Limited	Complex

3.4 Conclusion

Bus topology is much easier and cheaper to install than full mesh topology.

Mesh topology provides excellent reliability but requires considerably more cables, interfaces, configuration, and maintenance.

Therefore, full mesh is generally reserved for situations where reliability is more important than installation cost and simplicity.

4. HYBRID TOPOLOGY VS STAR TOPOLOGY – SCALABILITY

4.1 Hybrid Topology

Hybrid topology combines two or more different network topologies.

For example, a large organisation may use:

Mesh/Ring → Campus Backbone

Star → Building Network

Tree/Star → Department Network

This allows each section to use a topology that best matches its requirements.

4.2 Star Topology

Star topology is also scalable.

New devices can be connected to unused ports on the central switch.

However, the number of directly connected devices is limited by the available ports.

For example, a 24-port switch cannot directly accommodate more than its available ports without adding additional networking equipment.

4.3 Hybrid Scalability

Hybrid topology allows different network segments to be expanded independently.

For example, a university can add a new building by connecting a new building-level network to the existing campus backbone.

The internal topology of the new building does not have to be identical to the existing buildings.

4.4 Comparison

Feature	Hybrid	Star
Scalability	Very high	High
Flexibility	Very high	Moderate
Expansion	Segment-based	Switch-based
Complexity	Higher	Lower
Suitable for large networks	Yes	With additional hierarchy
Cost	Higher	Lower

4.5 Important Technical Point

A large enterprise network often uses multiple interconnected star networks in a hierarchical design.

Therefore, star topology itself can scale significantly when switches are interconnected.

Hybrid topology is considered more flexible because it can combine different topologies based on specific requirements.

4.6 Conclusion

Hybrid topology provides greater flexibility and scalability for large and complex organisations.

Star topology is simpler and is excellent for individual offices or buildings, while hybrid topology is more suitable when different sections of an organisation have different networking requirements.

5. MAINTENANCE COMPARISON

5.1 Star Topology

Star topology is relatively easy to maintain.

Because each device has an individual connection to the central switch, technicians can identify the affected device or cable easily.

For example, if one computer loses connectivity, the administrator can check:

1. Computer network adapter
2. Ethernet cable
3. Switch port
4. IP configuration

Other devices remain operational while troubleshooting takes place.

5.2 Mesh Topology

Mesh topology is highly reliable but more difficult to maintain.

The large number of connections means that administrators must monitor many links.

A problem may require checking multiple paths, interfaces, routing configurations, and network devices.

Therefore, maintenance requires skilled network administrators.

5.3 Bus Topology

Bus topology is relatively difficult to troubleshoot.

If the backbone cable has a fault, several or all devices may be affected.

The administrator must locate the exact position of the cable failure.

This can be time-consuming.

5.4 Hybrid Topology

Hybrid topology has moderate to high maintenance complexity.

The reason is that it combines different topologies and technologies.

For example, a university network may contain:

- Fiber backbone
- Layer 3 switches
- Building-level star networks
- Wireless access points
- VLANs
- Routers
- Firewalls

Therefore, administrators must understand multiple technologies.

5.5 Maintenance Comparison

Topology	Maintenance Difficulty	Main Reason
Star	Low	Individual links are easy to isolate
Bus	Moderate/High	Backbone faults affect many devices
Mesh	High	Large number of connections
Hybrid	Moderate/High	Multiple technologies and topologies

5.6 Overall Conclusion

Star topology is generally the easiest to maintain.

Mesh topology provides the highest redundancy but requires more complex administration.

Bus topology is simple in structure but backbone faults can be difficult to locate.

Hybrid topology provides flexibility and scalability but requires skilled administration because several network designs may exist together.

6. OVERALL COMPARATIVE TABLE

Parameter	Star	Bus	Mesh	Hybrid
Cost	Medium	Low	Very High	High
Performance	High	Low/Medium	Very High	High
Reliability	High	Low	Very High	High
Fault tolerance	Medium	Low	Very High	High
Installation	Easy	Easy	Difficult	Moderate/Difficult
Maintenance	Easy	Difficult	Difficult	Moderate
Scalability	High	Low	Complex	Very High
Cabling	Medium/High	Low	Very High	Variable
Complexity	Low	Low	Very High	High
Modern usage	Very common	Rare	Critical networks	Enterprise/Campus